

Architectural Particular Specification

QUALITY ASSURANCE

SECTION 1.6

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SECTION 1.6

QUALITY ASSURANCE

PART 1 GENERAL

1.01 SUMMARY

The contractor shall describe in his tender and implement during the contract a quality system to verify that the whole of the works comply with the requirements of the contract.

The quality system shall be equivalent to BS 5750 / ISO 9002 and a quality plan shall be submitted by the contractor prior to the commencement of the works. The quality plan shall fully describe the specific practices, resources and activities for the implementation of the quality system on the contract. The quality plan shall include provisions and procedures under the following headings :-

A. ORGANIZATION

1. The Contractor shall allocate to a senior member of his site team the responsibility and resources necessary for ensuring that the quality system is effective. The name of this Quality Control Engineer and his C.V. must be submitted for the Architect's approval prior to any construction works commence.
2. The Contractor shall provide details and duties of the personnel to be involved and any independent inspecting authorities which he proposes to employ.
3. The Contractor shall ensure that all personnel have appropriate qualifications, experience or training for the tasks allocated to them.

B. REVIEW OF QUALITY SYSTEM

1. The Contractor shall establish a Quality Assurance Committee which includes members from :-
 - i The Contractor including the Project Manager and the Quality Control Engineer;
 - ii The Client's Project Manager, Quality Manager and their site representatives;
 - iii The Architect and / or the Consultants and their site supervisory staff.
2. The committee shall execute a quality audit and carry out regular meetings on site to ensure the Contractor's management / worker responsibilities and methods of operation are adequate, quality meets with the specifications and corrective actions are taken. Other parties such as the sub-contractors may be called upon to attend the meeting where required but will not be considered as part of the committee.
3. The Contractor shall submit report to the committee during each meeting.

C. DOCUMENTATION

1. The Contractor shall maintain records to substantiate conformance to specified requirements, including work carried out by sub-contractors, manufacturers, suppliers, and independent inspecting authorities.
2. Any records which indicate that material or workmanship in any part of the works does not comply with the specified requirements shall be submitted without delay together with the Contractor's proposal for rectification.

3. All records shall be retained on site and made available for examination.
4. The Contractor shall establish and maintain procedures to control all documents and data for each quality plan activity.
5. The Contractor shall keep accurately dated records of the progress of the works.

D. SUBMISSION OF SAMPLES

1. Ensure that all materials, equipment and services including those provided by the sub-contractors, nominated sub-contractors and nominated suppliers conform to the specification, are properly packaged and that the “shelf life” has not been exceeded.
2. Prepare purchasing documents to contain clear description of the materials, equipment services ordered during the dates for and form of delivery.
3. Ensure that incoming materials and equipments are checked on delivery, properly stored, protected and are not used or processed until inspected or otherwise verified as complying with the various specifications.
4. Establish procedures for identifying, controlling, segregation, recording and disposing of materials, equipment.
5. Establish a sample procedure for the procurement of, submission to, and endorsement by the Architect.
6. Provide a monitoring process to ensure all subsequent materials, workmanship, etc. shall be to the standard of the endorsed samples.

E. CONTROL OF SUB-CONTRACTORS

1. The quality system shall include for all works executed by sub-contractors and the Contractor shall ensure that all the sub-contractors operates the necessary procedures and controls required by the contract.

F. PROVISION OF ‘MOCK-UPS’

This part shall be read in conjunction with the Specification – Preliminaries and the appendix for the requirements for prototypes and trial areas. Construct ‘Mock-ups’ of sample units / sample areas as provided for the contract to meet the following requirements :-

1. Materials, workmanship, etc. are to be to the standard endorsed by the Architect as a sample.
2. Each stage of the ‘Mock-ups’ shall be inspected by the Architect and there shall be no areas covered up before being endorsed by the Architect.
3. Once endorsed by the Architect, the quality of the materials, workmanship, etc. in the ‘Mock-ups’ shall become the standard of all subsequent work.
4. The timing for constructing ‘Mock-ups’ shall be shown on the Construction Programme.
5. All materials and fittings for the ‘Mock-ups’ shall be made available on time.
6. Arrangements are to be made for preserving and protecting the completed ‘Mock-ups’.

G. QUALITY PLAN FOR INDIVIDUAL TRADES OF WORKS

1. Prior to work execution of individual trades of work, the Contractor should liaise with their various sub-contractors in submitting a Quality Plan for each work trade for Architect's consideration / vetting. Such quality plan should at a minimum cover the following topics (which must never be construed as exhaustive) and the Architect is at liberty to request further information to be incorporated into such plan:-
2. Procedures for material delivery from, say, off-site factory to site and subsequently to the works area at which they are installed / applied.
3. Protection of the materials / products at various stages of works from the time when they are dispatched from factory / workshop till the handover of the completed works to the Employer.
4. Interfacing of the works with other work trades, and the means / solution to resolve any possible interfacing problems.
5. Method statement and detailed working procedures for the installation / construction / application of the work trade. The interfacing with associated work trades should also be duly addressed.
6. Detailed procedures for Quality Control at each work stage, and the remedial measures to be adopted when any of the specification requirements / pre-determined standard or design parameters cannot be achieved.
7. Format of Inspection Form(s) together with the list of items / work stages to be inspected list of items / work stages to be inspected by Architect, Engineer, Building Services Engineer and / or Owner's Site Representative as the case may be.
8. Construction Tolerances anticipated, which shall be in line with those indicated in the specification / drawings of the contract document.
9. For avoidance of doubt, the individual work trades requiring submission of Quality Plan should also cover those works which are to be performed by any direct and / or nominated sub-contractor(s) appointed, and it is the Main Contractor's sole responsibility to coordinate with them for preparing and submission of the requisite Quality Plan.

H. CONSTRUCTION CONTROL

1. Work Procedures
 - i Prior to the commencement of each work activities, the Contractor shall perform a 'sample work procedure' for the work party to agree. The work party shall include the Quality Control Engineer, the site agent, the site foremen and their assistants, the subcontractor for the particular work, the Resident Engineer, the COW and / or the BS Engineer or any responsible persons for the work where necessary. The 'sample work procedure' shall conform to the specification and agreed by the Resident Engineer, the COW and / or the BS Engineer prior to actual work commence.
 - ii Any materials, mixture of materials or admixtures being used is agreed and to be used in all subsequent work shall be in right proportion conformed to the agreed work procedure.
 - iii Once the work procedure is agreed, the work procedure shall become the standard procedure for all subsequent work.
 - iv The work procedure shall be documented with record photos by the Contractor and endorsed by the Architect or his representative prior to distribution.
2. Process Control

- i The Contractor shall plan, monitor and control construction processes which directly affect quality.
 - ii The Contractor shall :-
 - a. Ensure that construction activities are carried out under controlled conditions with adequate supervision and in accordance with the various specifications.
 - b. Ensure that both on and off site storage is suitable for the materials being stored and that the quality is preserved.
 - c. Ensure that material handling is carried out carefully and efficiently and that the quality is preserved.
 - d. Establish the criteria for workmanship by written instructions, or photographs or by representative samples.
 - e. Care for and protect installed materials and completed work from damage or deterioration.
3. In-process Inspection and Testing
- i The Contractor shall define the procedure in performing in-process inspection and testing and in releasing completed works to their next construction stage.
 - ii This procedure shall cover the inspection and testing of those works which are subsequently covered permanently by and forming an integral part of other elements of the works.
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 - a. Setting out
 - b. Wall-tie holes for external wall
 - c. Floor screeding
 - d. Floor tiling
 - e. Spatterdash
 - f. Window frame installation
 - g. Window frame grouting
 - h. Waterproofing
 - i. Timber door frame installation (including sub-frame)
 - j. Metal door frame installation
 - k. Bath tub / Shower tray installation
 - l. Blockwork / Brickwork
 - m. Wall plastering
 - n. Wall tiling
 - o. Ceiling plastering
 - p. Gas sleeve opening
 - q. Construction of shoring and strutting, temporary support
 - r. Concrete inserts e.g. anchor bolt, bracket, channel
 - s. Pipe ducts
 - t. Others

4. Non-conforming Work

- i Where in the opinion of the Architect, any of the finished work or materials or workmanship in any part of the Works fails to comply with the Specification that part of the Works will not be accepted.
- ii Any work which the Architect judges to be inferior in respect to an approved sample or trial panel or to be unacceptably different from parts of the Works already constructed or which is subsequently stained or damaged will not be accepted.
- iii All such work shall be cut out and removed from the site and replaced or otherwise dealt with in an approved manner.

5. Corrective Action

Establish and maintain a documented procedure providing for :-

- i A continuing analysis of defects and their correction.
- ii A continuing process for inspecting and identifying non-conforming materials and equipment to prevent them from being installed.
- iii An assurance that such corrective action is effective.

6. Protection of the Works

- i The Contractor shall be responsible for controls on handling, storage, preservation, protection and delivery of materials and works to reduce risk of damage or deterioration. The Contractor shall formulate procedures for the followings :-
 - a. Handling - operatives are instructed in the care they must taken when handling materials and finished works to prevent damage or deterioration.
 - b. Storage and Preservation - storage areas and designated for material pending use. To prevent damage or deterioration, the condition of material in storage are assessed regularly.
 - c. Protection and Delivery - materials received are generally retained in their protective packing, off the ground and segregated whenever applicable.
- ii Finished works are protected in accordance with contractual requirements or trade practices to prevent them from damage after final inspection and testing.

7. Completed Work Inspection and Testing

- i Final inspection and testing is performed on completed works prior to handover to the client. The final inspection activities include a review of all previous rectification records and the final checking of the completed work by using applicable instruction or checklists.
- ii The Contractor shall establish the procedures for such inspection and testing. Any nonconformities found are rectified and re-inspected to ensure compliance with the specified requirements.

I. LABORATORY TESTING

1. SELECTION AND PAYMENT

- i Materials testing shall only be carried out by laboratories holding HOKLAS accreditation for the relevant tests or operating systems in accordance with ISO/IEC Guide 25. Tests shall be carried out in accordance with the relevant product/materials standards & Codes of Practices as referenced in the Specifications.
- ii Testing Equipment: Calibrated at reasonable intervals with devices of an accuracy traceable to CSI: 1990 for concrete testing and other relevant Standards.

- iii The Contractor shall employ and pay for services of an independent testing laboratory to perform specified inspection and testing. The selected independent testing laboratory shall be approved by the Architect (and licensed, if required by the local Building Authorities).
- iv Employment of testing laboratory shall in no way release the Contractor of obligation to perform work in accordance with requirements of Contract Documents.
- v The Contractor may not employ the same testing laboratory engaged by the Employer, unless otherwise agreed in writing by the Employer.

2. SUBMITTALS

The Contractor shall submit the followings:

- i Submittals:
 - a. Testing laboratory name, address, fax and telephone numbers, and names of full time Registered Structural Engineer and responsible officer.
 - b. Copy of report of laboratory facilities inspection made by laboratory accreditation body during most recent inspection, with memorandum of remedies of any deficiencies reported by the inspection.
 - c. All inspection and testing reports referred in the Contract Documents.
- ii Submit to:
 - a. Project Manager
 - b. Architect
 - c. Structural and Geotechnical Engineer
 - d. Building Services Engineer
 - e. Sustainability Consultant
 - f. Security Consultant
 - g. Acoustic Consultant.
 - h. Landscape Consultant
 - i. Buildings Department or other Statutory Authority, when required.

3. DELIVERY STORAGE & HANDLING

- i Deliver to laboratory, at designated location, adequate samples of materials proposed to be used which require testing, along with proposed mix designs. A representative of the testing laboratory or the Architect is to be present when samples of any material are taken for testing. The Contractor shall inform the Architect, and his site representative, in writing, minimum 24 hours notice of such sampling.
- ii Cooperate with laboratory personnel, and provide access to the Project Site and to manufacturer's facilities.
- iii Provide incidental labour and facilities to provide access to work to be tested, to obtain and handle samples at the Project Site or at source of products to be tested, to facilitate tests and inspections, storage and curing of test samples. Materials being tested shall be of identical to that being incorporated into the Works.
- iv Notify the Architect and testing agency 48 hours minimum prior to expected time for operations requiring inspection and testing services.

4. LABORATORY INSPECTION AND TESTING RESPONSIBILITIES

- i The testing laboratory shall perform inspections, tests and other services specified in the Contract Documents.

- a. The testing laboratory shall provide the required qualified personnel at the Project Site.
 - b. The testing laboratory shall cooperate fully with the Architect and Contractor in the performance of services.
- ii Laboratory Reports shall be promptly submitted to the Contractor by the testing laboratory in accordance with Section 1.5 – Submittals Procedures indicating observations and results of tests and indicating compliance or non-compliance with the Contract Documents.
- iii Furnish samples of materials, design mixes, equipment, tools, storage and assistance as required.
- iv Retesting due to non-conformance with the Contract Documents shall be performed by the same testing agency upon instructions from the Architect. The Contractor shall be responsible for payment for all retesting operations.
- v Testing laboratories shall attend pre-construction conferences and progress meetings.
- vi Laboratory Reports:
 - a. After each inspection and test, the Contractor shall promptly submit 3 copies of the laboratory report to the Architect, Structural Engineer and Project Manager, unless otherwise noted.
 - b. Include:
 - Date issued.
 - Project title and number.
 - Name of inspector.
 - Date and time of sampling or inspection.
 - Identification of product and Specifications Section.
 - Location in the Project.
 - Type of inspection or test.
 - Date of test Environmental conditions.
 - Results of tests.
 - Conformance with Contract Documents.
- vii When requested by Architect, provide interpretation of test results.
- viii Limits on Testing Laboratory Authority :
 - a. Laboratory may not release, revoke, alter, or enlarge on requirements of Contract Documents.
 - b. Laboratory may not approve or accept and portion of the Work.
 - c. Laboratory may not assume any duties of the Contractor.
 - d. Laboratory has no authority to stop the Works.

PART 2 PRODUCTS

(NOT USED)

PART 3 EXECUTIONS

3.01 INSPECTIONS AND TESTS

- A. Assemble all inspection and test reports into the “Close-Out Documents Book” as part of the Project Close-Out procedures stated in Section 1.14.
- B. Perform all inspections and testing as specified in the Contract Documents, and as required by the Building Authorities.

END OF SECTION